Introduction to Machine Learning

Lecture 2

25 January 2022

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Outline

- Overview of the last class
- Basic Terminologies widely used in ML
- Types of Machine Learning

Introduction

- Machine Learning
  - Grew out of work in AI
  - New capability for computers

- ML has revolutionized the way we live in a bigger way

- Some Real time Examples which are part of our daily lives

Computer Vision (Cameras, Object detection, Recognition), Natural language processing, handwriting recognition, Self driving cars, …….

What is Machine Learning ?

Definition

Arthur Samuel (1959). Machine Learning:

Field of study that gives computers the ability to learn without being explicitly programmed.

Tom Mitchell (1998). Well-posed Learning Problem:

A computer program is said to *learn* from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.

Kevin Murphy (2012):

Machine learning is the study of computer algorithms that improve automatically through experience.

"A computer program is said to *learn* from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E."

Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. Please categorize the following ?

a. Classifying emails as spam or not spam.

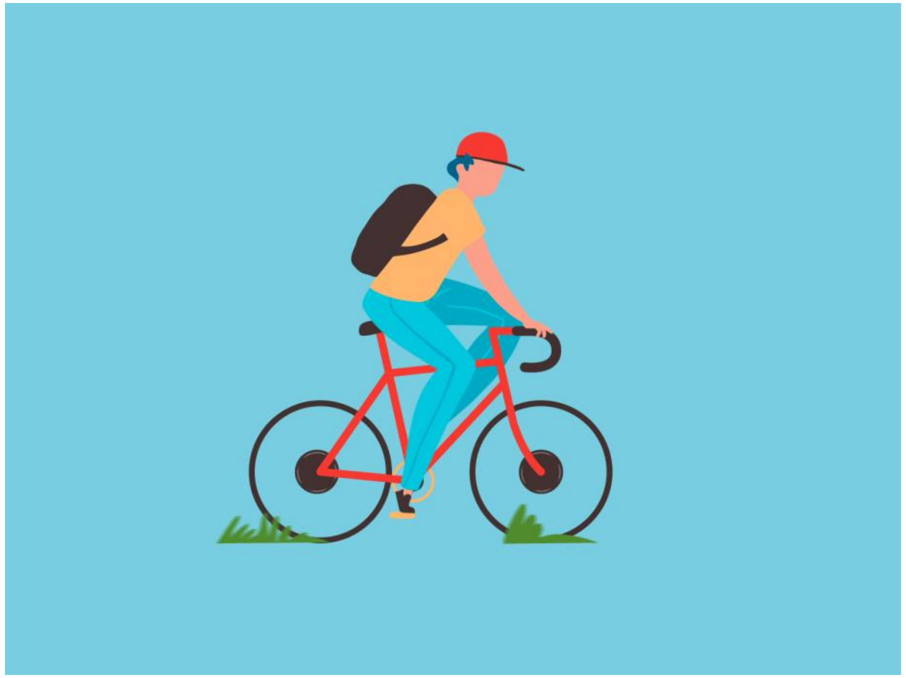
b. Watching you label emails as spam or not spam.

c. The number (or fraction) of emails correctly classified as spam/not spam.

True Challenge to ML

- Is to solve the tasks that are easy for people to perform but hard for people to describe formally

Can some one tell the example ???



What ML Consists of ?

- Statistics, Optimization, Probability, Information Theory, Psychology, Philosophy, Neurobiology, Control Theory, Computer Science, Programming, Calculus, Linear Algebra, etc.

When Machine Learning is feasible ?

What we need/ or when can we apply ML ?

1. We need Data. {Without Data ML can not be applied}

Data can be: Numbers, Text, Image, Audio, Speech, etc.

What else we can say about the data ? Or What you expect from the data ?

2. The Data Should follow some pattern

Patterns means– It should be mathematically expressible

Pattern is there, what's next ?

Size in feet^2 (x)	Price ($) in 1000's (y)
2104	460
1416	232
1534	315
852	178
…	…

Features and Label

Label- Ground truth/output

Features-Inputs/variables

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	yes
D6	Rain	cool	Normal	Strong	Yes
D7	Overcast	Cool	High	Weak	No
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Person playing tennis on a particular day based om weather conditions

Lets Check out with some other data sets

Lets see what is there in the Data Sets:

- House Price Prediction
- Iris Data Set
- Wisconsin Data set

Repositories to find the Data Sets

https://www.kaggle.com/datasets

https://archive.ics.uci.edu/ml/index.php

Machine Learning Terminologies

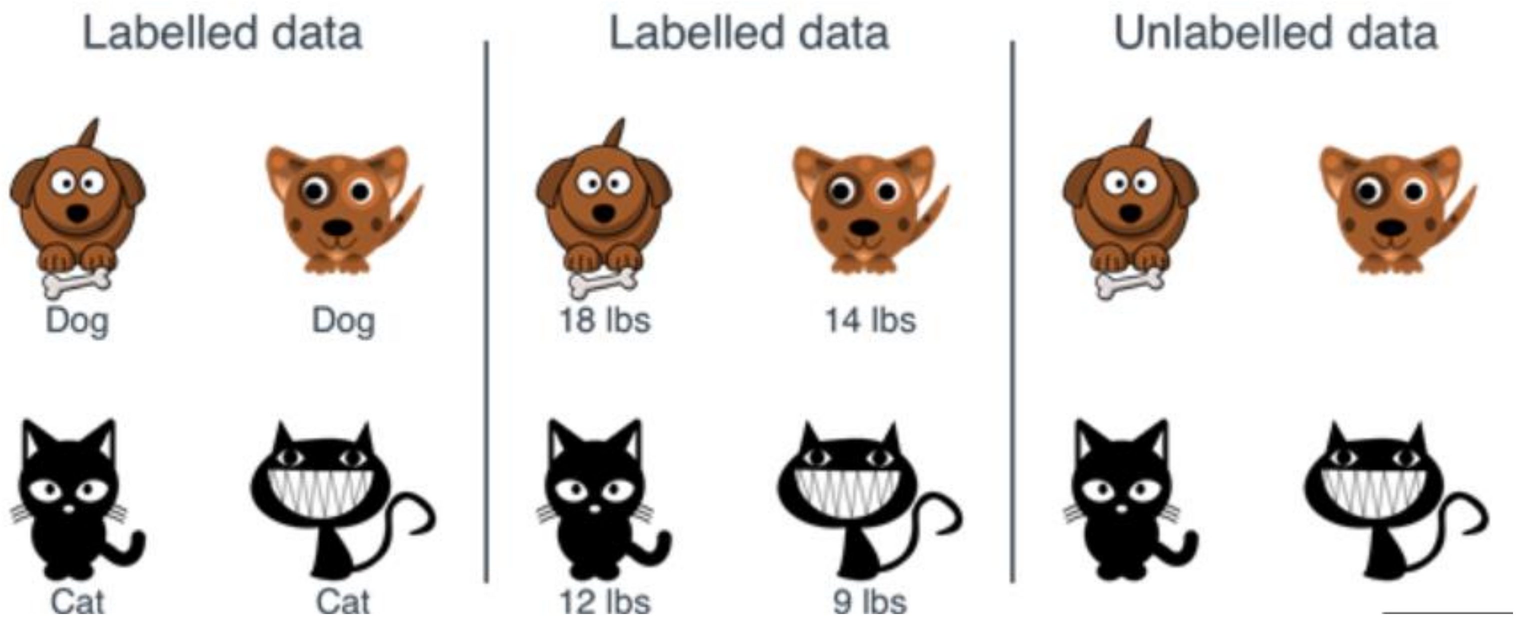
- Labelled Data
- Un-labelled Data
- Classification
- Regression
- Clustering
- Train Data
- Test Data
- Features

Labelled Data

Data Comes with Label

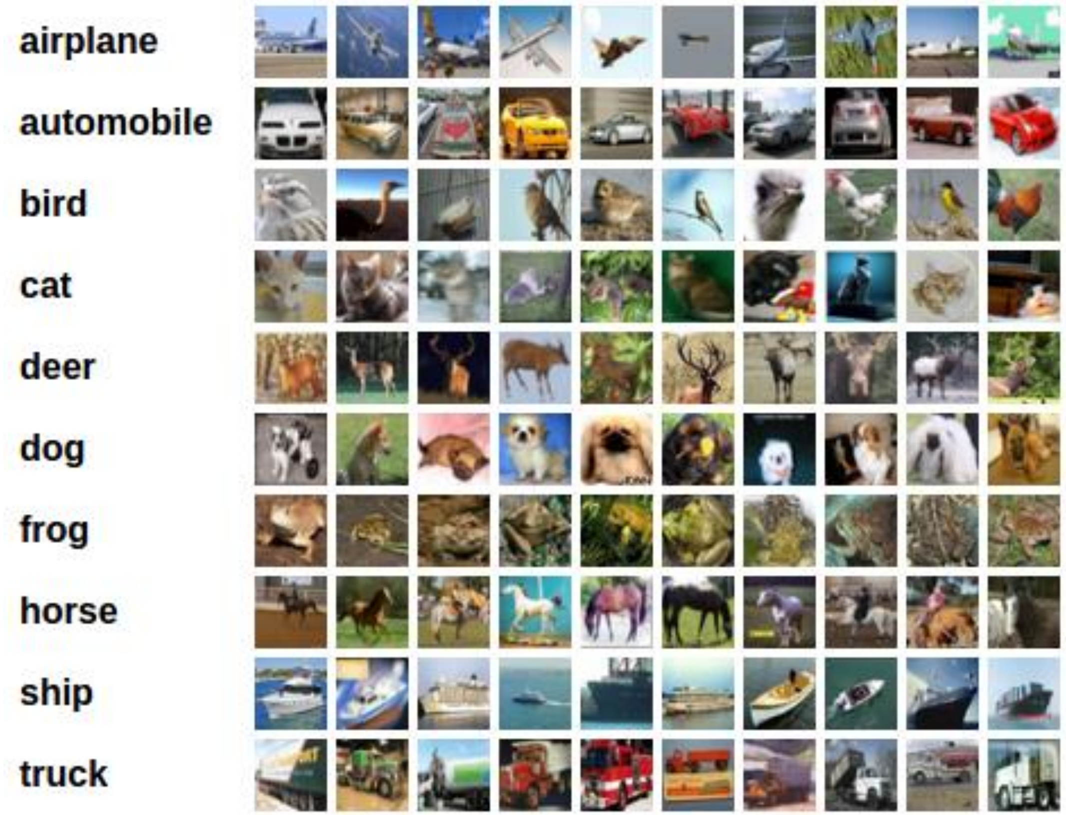
The label can be name, number, Boolean (Yes or No), etc.

- **Labeled data** is a group of samples that have been tagged with one or more labels.
- Labeling typically takes a set of unlabeled data and augments each piece of it with informative tags.
- For example, a data label might indicate whether a photo contains a horse or a cow, which words were uttered in an audio recording, what type of action is being performed in a video



Source: Kaggle

Few Examples



Action recognition

How about Voice/Speech labelling ?

Unlabeled Data

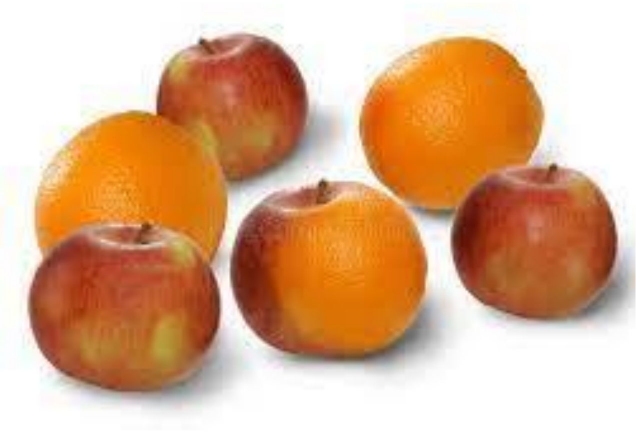
Unlabeled data is a designation for pieces of data that have not been tagged with labels identifying characteristics, properties or classifications

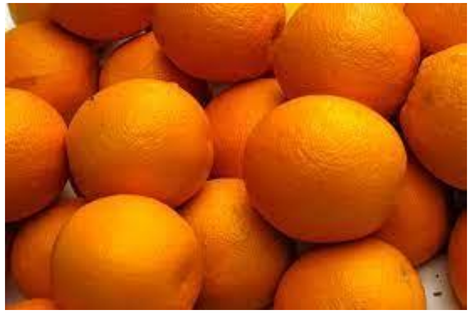
For example:

If I want to develop an automated algorithm for detecting the tumor in CT image.

If we approach the Hospital, they will give all the images without any labels, then we call it as unlabeled data.

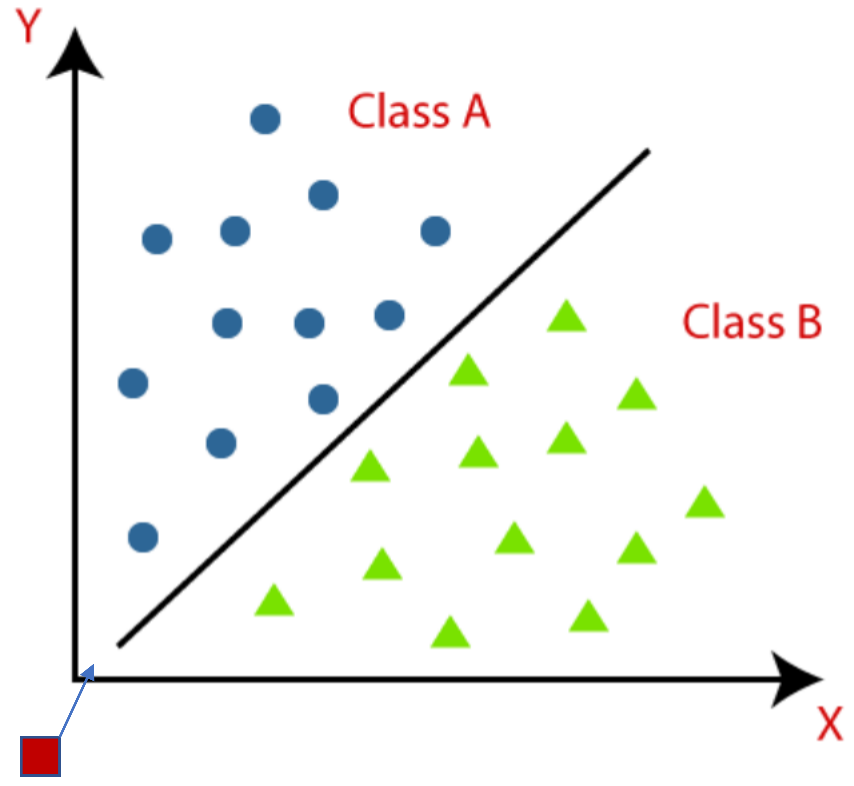
If the hospital, gave us the medical images along with its diagnosis report, then we call it as an labelled data.





Classification

- In Classification, a program learns from the given dataset or observations and then classifies new observation into a number of classes or groups. Such as, Yes or No, 0 or 1, Spam or Not Spam, cat or dog, etc.
- Classes can be called as targets/labels or categories.



Regression

- If algorithm wants to predict a continuous variable, then we call it as an Regression Problem.

Ex: Predicting the price of a house based on cost. Predicting how many items will be sold in a shop in a given day.

Which is below falls under Regression problem: ?

- Predicting age of a person
- Predicting nationality of a person
- Predicting whether stock price of a company will increase tomorrow
- Predicting whether a document is related to sighting of UFOs?

Train Data, Test Data, Features

To evaluate the algorithm. Datasets, are divided into two categories

Training datasets and Test data sets

Algorithms are trained on Training Data, and it is evaluated on Test Data.

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	yes
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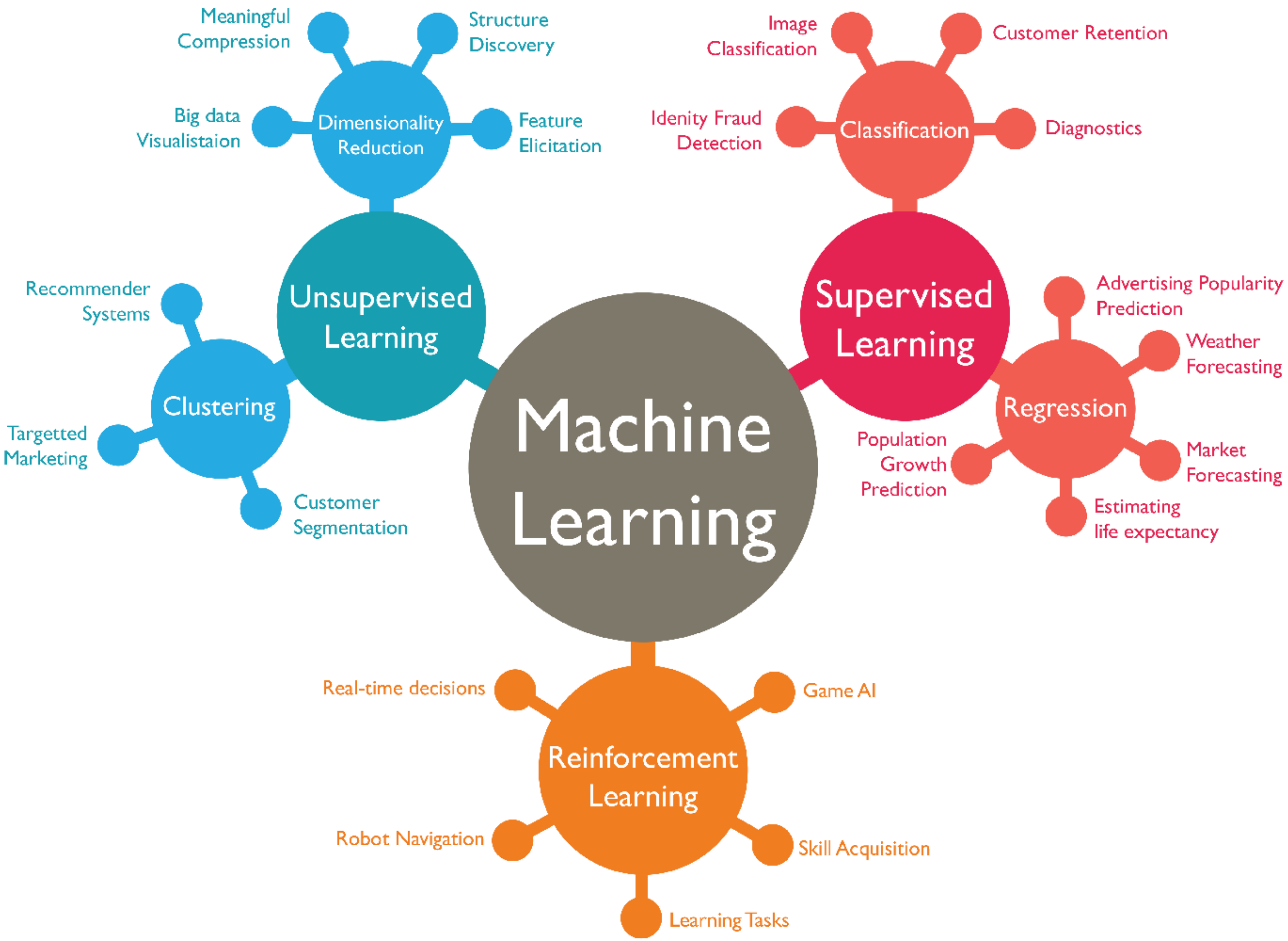
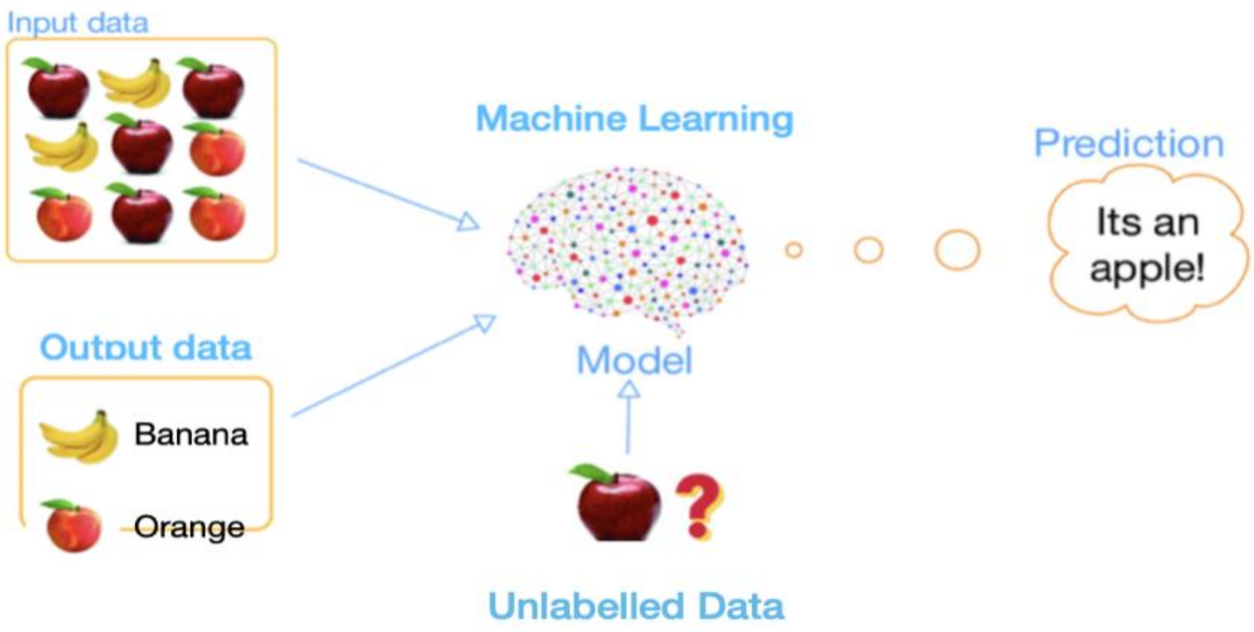
Different Types of Machine Learning

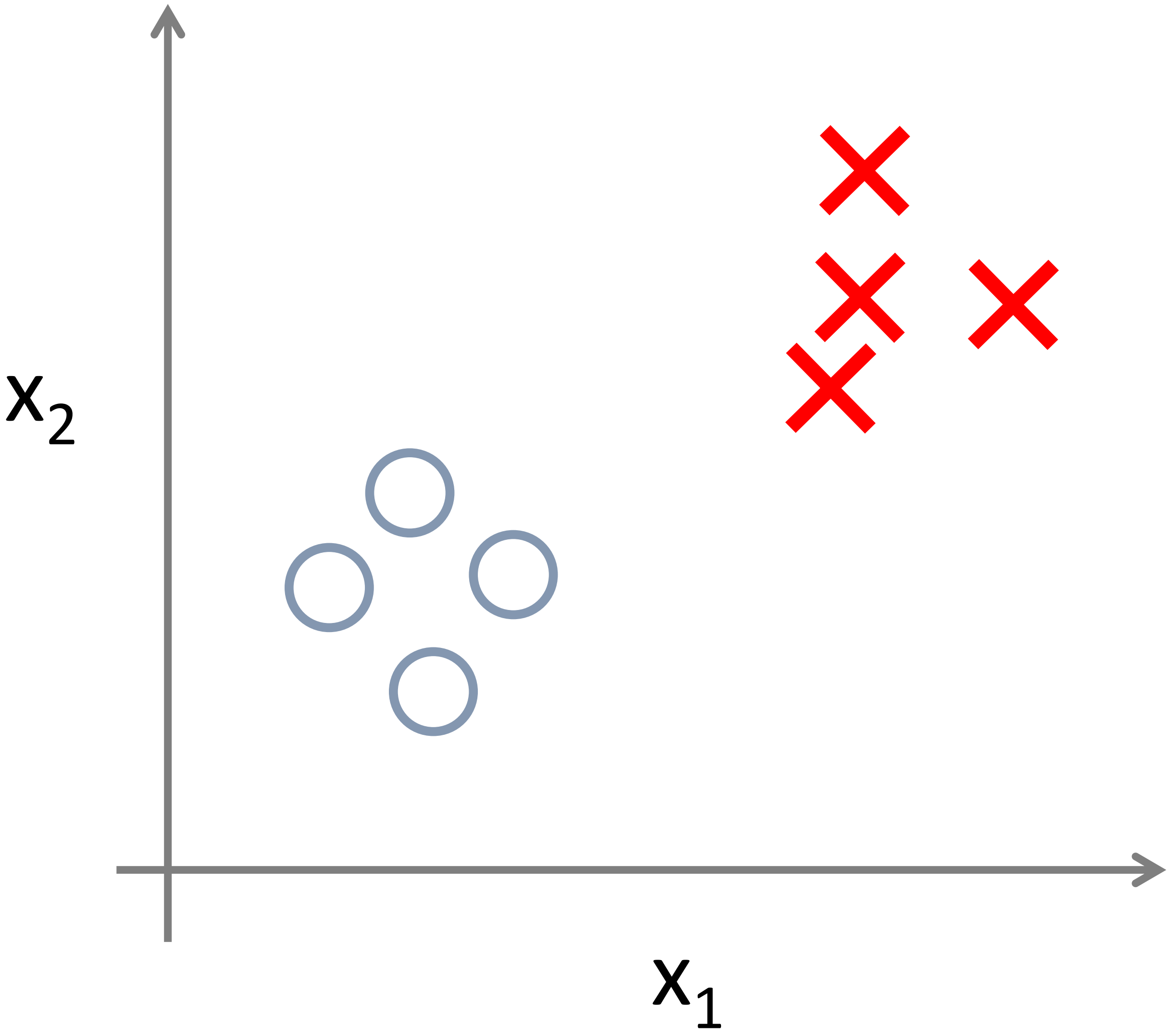


Supervised Learning

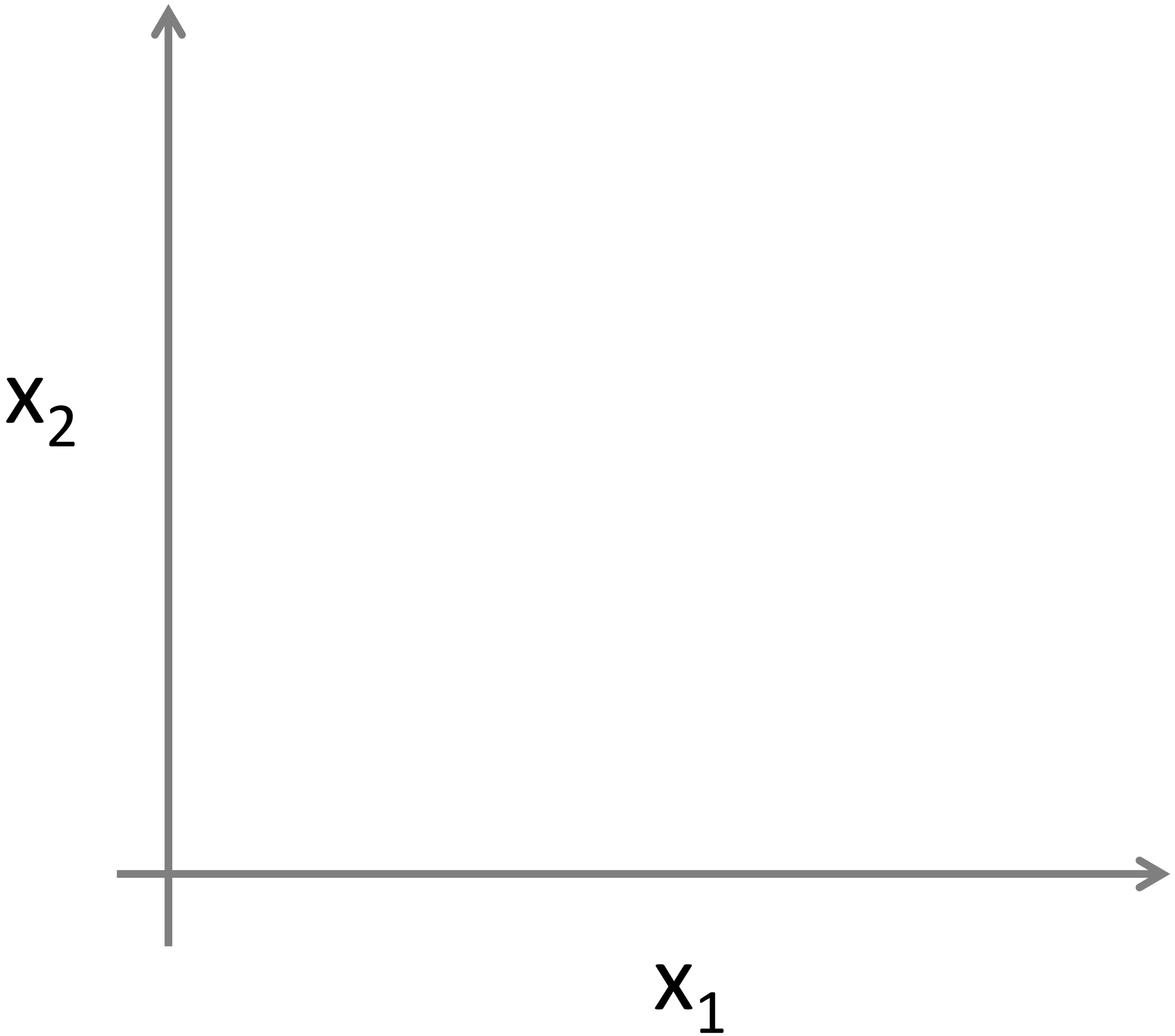
Supervised learning is the types of machine learning in which machines are trained using well "labelled" training data.



Supervised Learning- Classification



Supervised Learning- Regression

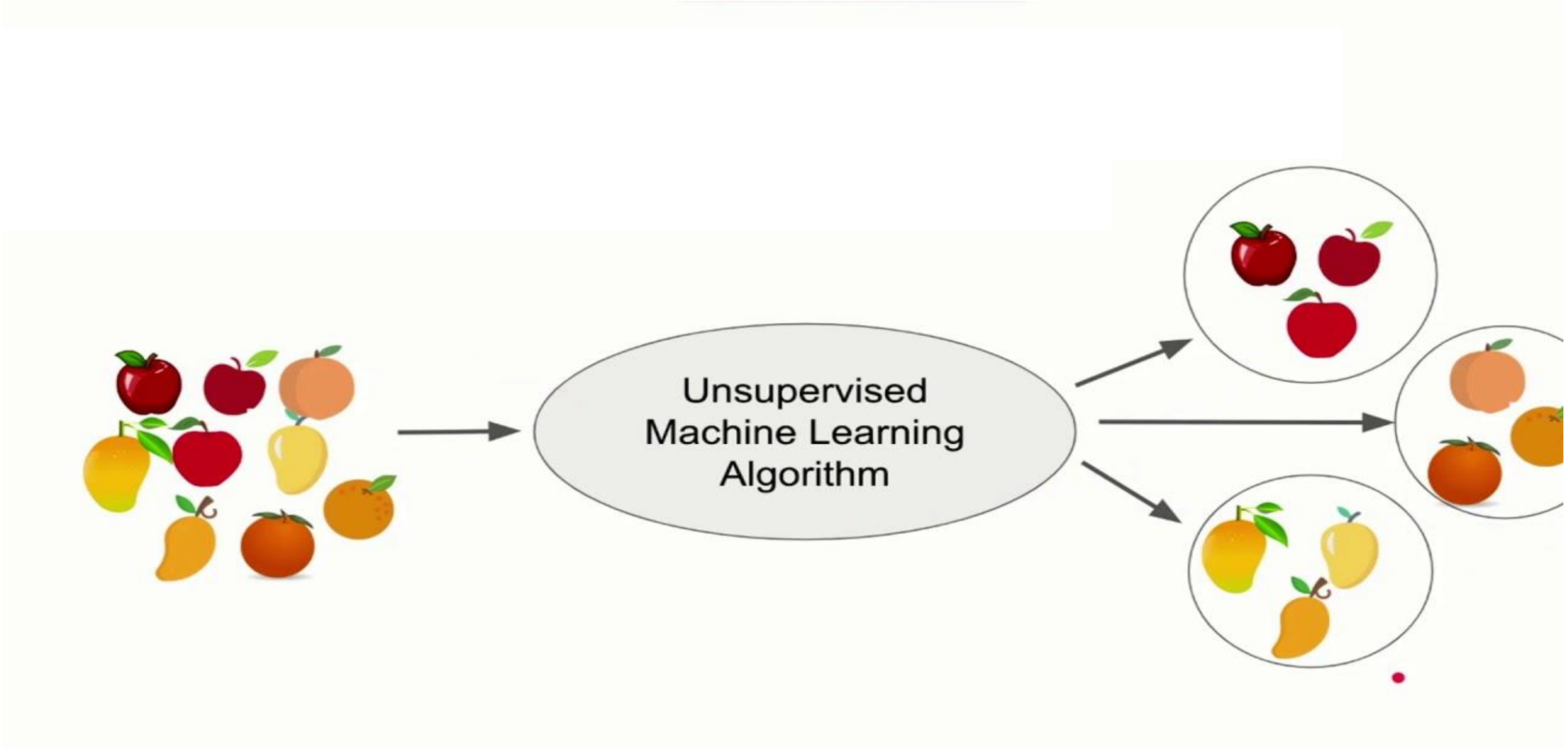


Regression Examples

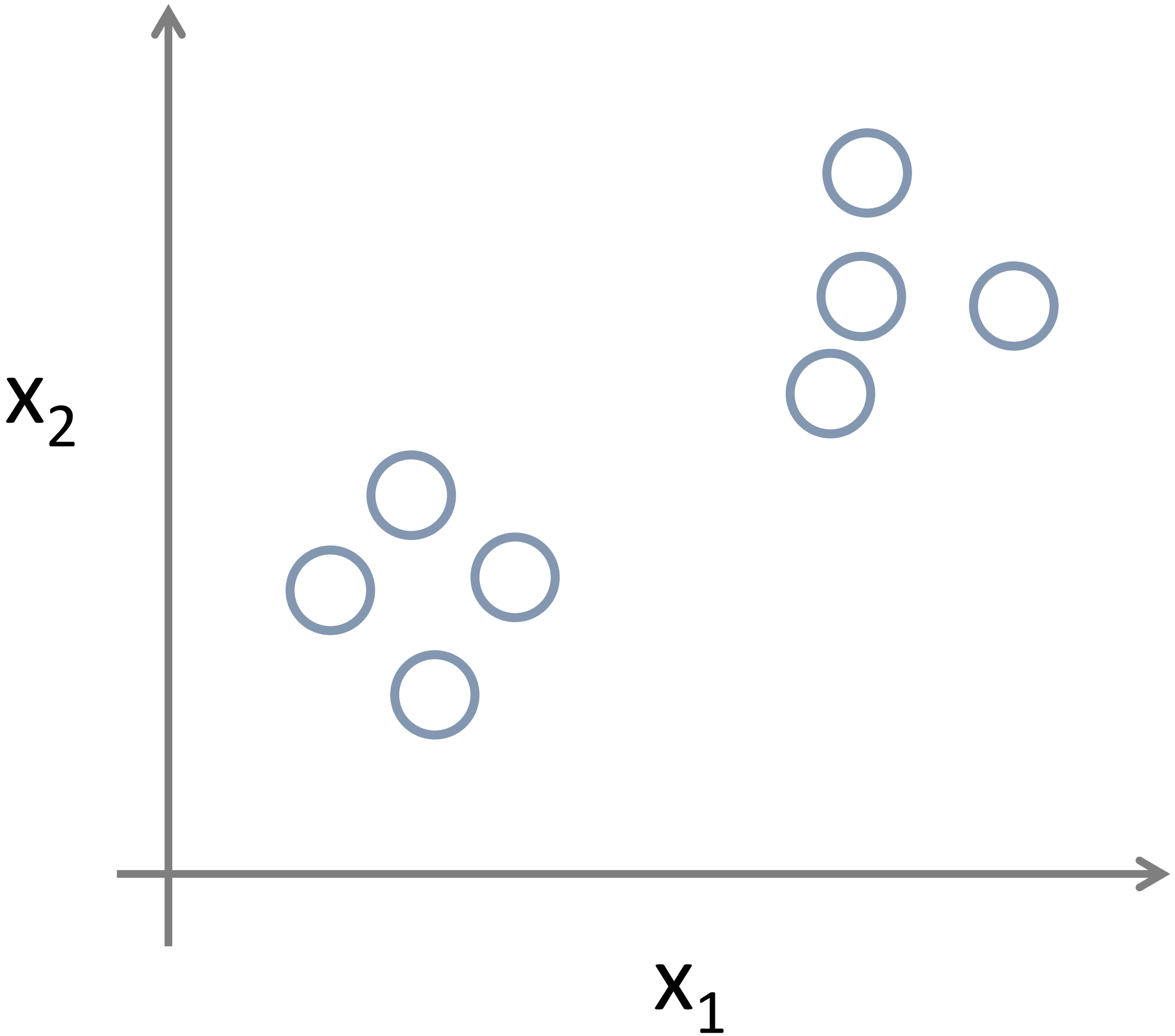
- Predicting the Stock Price
- Predicting the Temperature
- Predicting the Sugar levels
- Predicting the Gold Price
- Predicting the Projected score in a cricket match

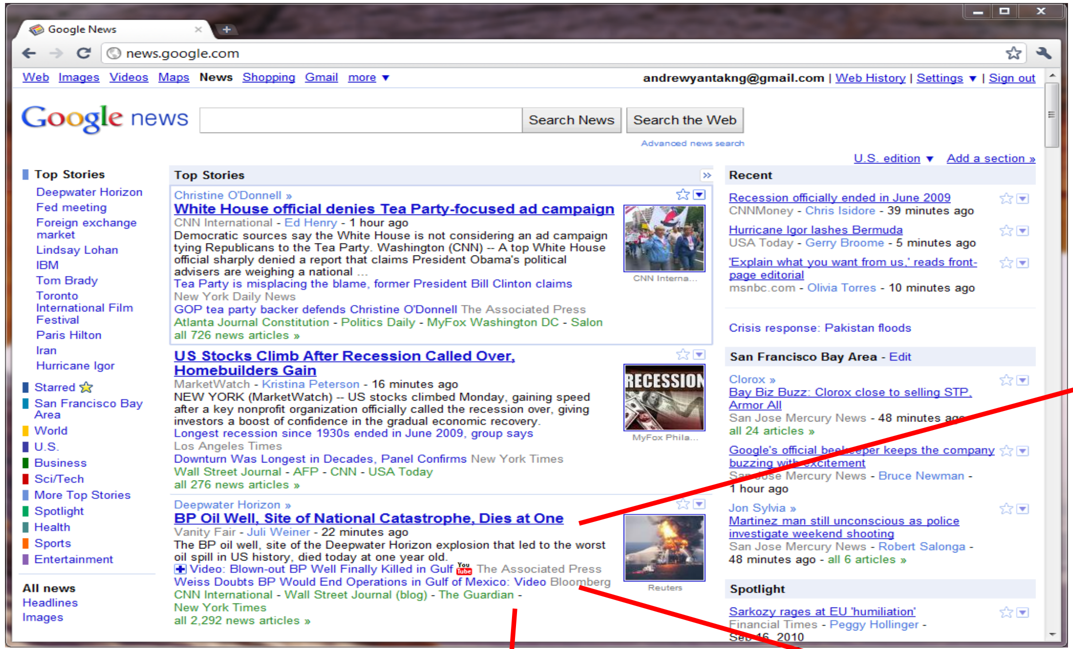
Unsupervised Learning

- Unsupervised learning is a type of [machine learning] in which the algorithm is not provided with any pre-assigned labels or scores for the training data. As a result, unsupervised learning algorithms must first self-discover any naturally occurring patterns in that training data set.



Unsupervised Learning

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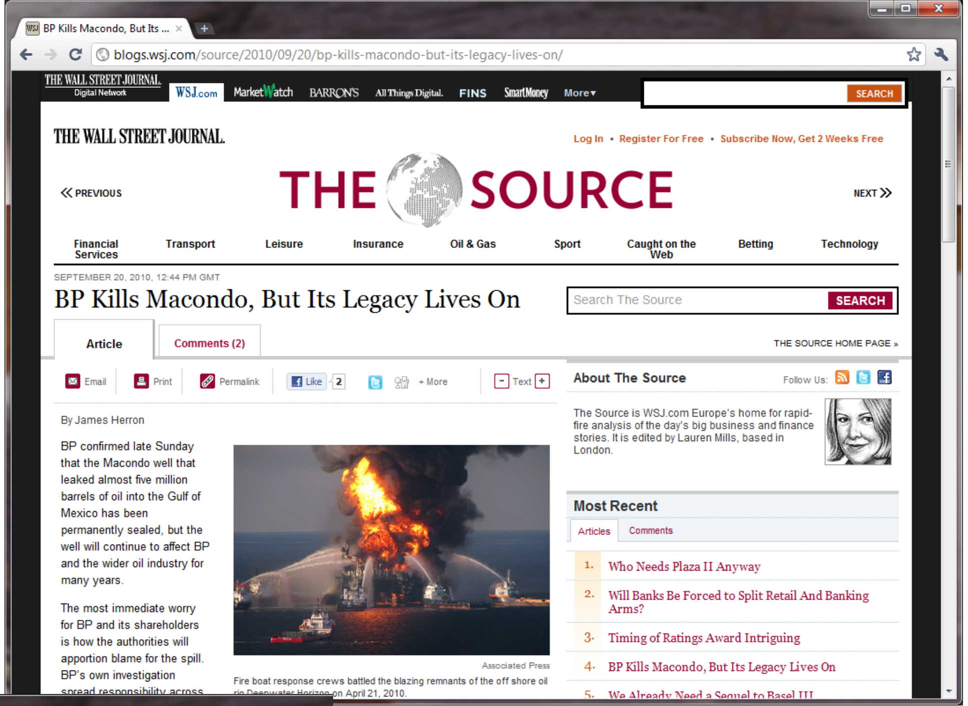
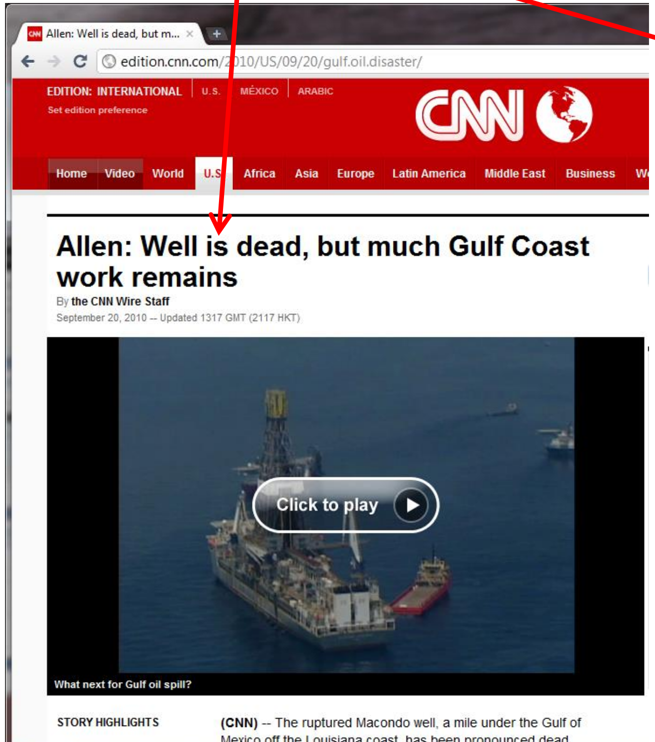

THE WALL STREET JOURNAL.
THE SOURCE
SEPTEMBER 20, 2010, 12:44 PM GMT
BP Kills Macondo, But Its Legacy Lives On
By James Herron
BP confirmed late Sunday that the Macondo well that leaked almost five million barrels of oil into the Gulf of Mexico has been permanently sealed, but the well will continue to affect BP and the wider oil industry for many years.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Allen: Well is dead, but much Gulf Coast work remains
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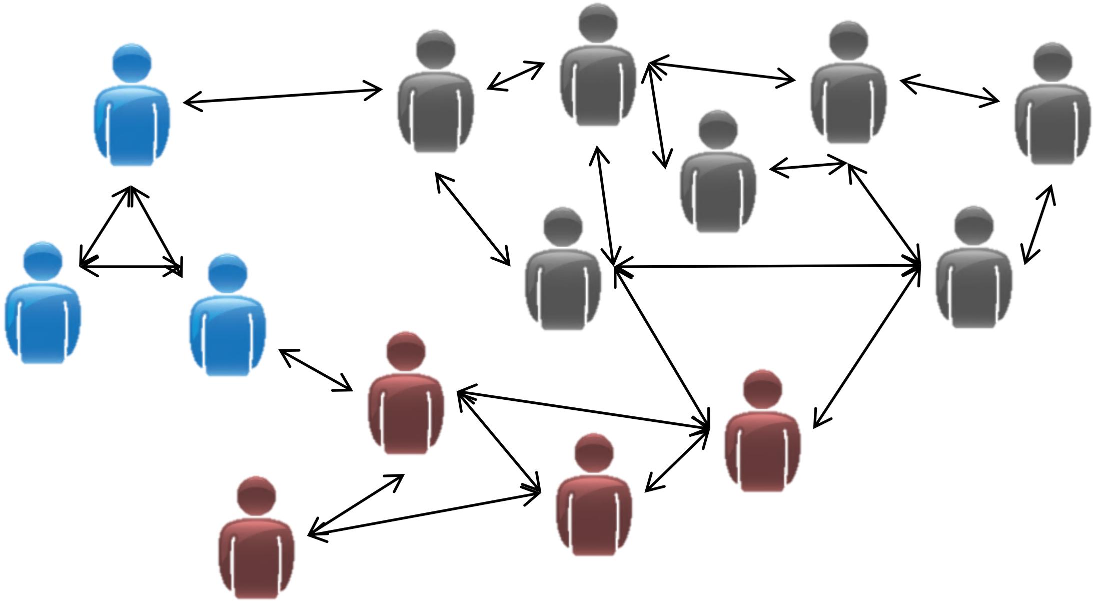
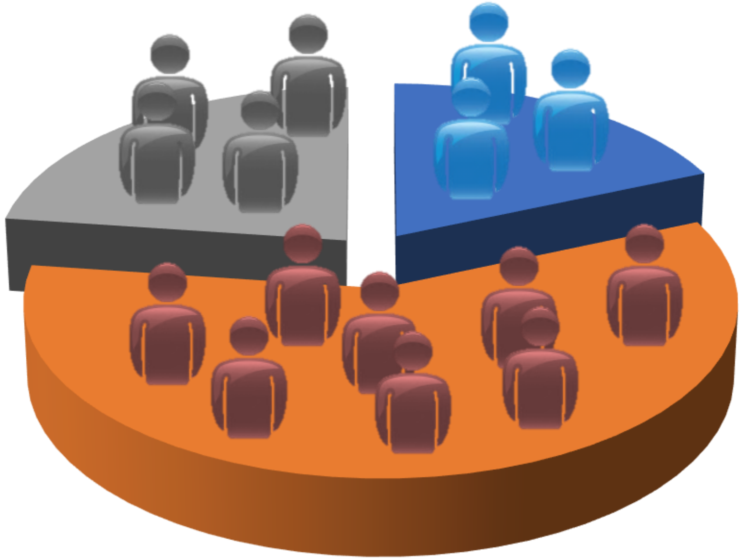
What next for Gulf oil spill?
STORY HIGHLIGHTS


guardian.co.uk
BP oil spill cost hits nearly $10bn
BP has set up a $20bn compensation fund after the Deepwater Horizon disaster, which has so far paid out 19,000 claims totalling more than $240m
Julia Kollewe
guardian.co.uk, Monday 20 September 2010 08.33 BST
BP's costs for the Deepwater Horizon disaster have hit $10bn. Photograph: Ho/Reuters

Organize computing clusters



Social network analysis



Market segmentation

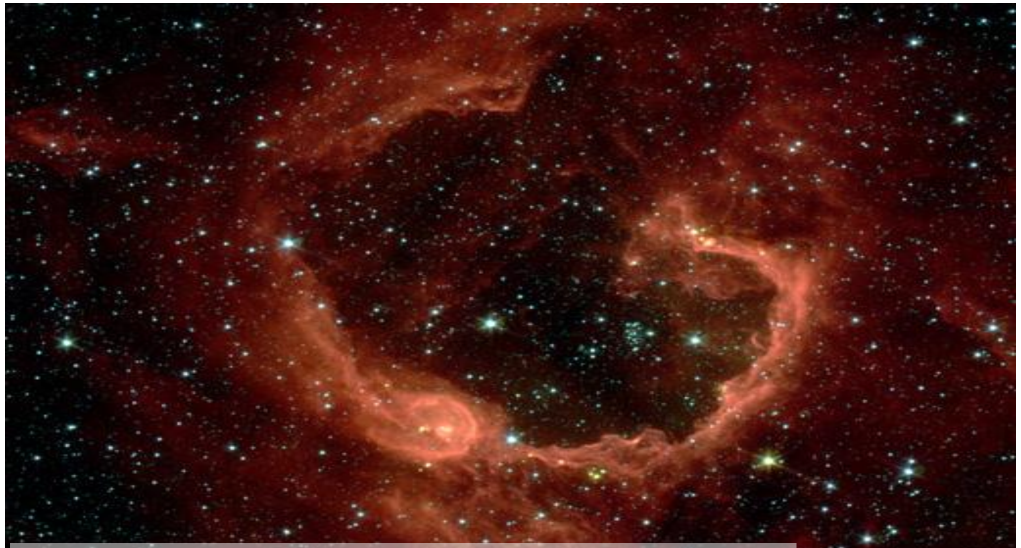


Image credit: NASA/JPL-Caltech/E. Churchwell (Univ. of Wisconsin, Madison)

Astronomical data analysis

Reinforcement Learning

- Reinforcement learning is an area of machine learning concerned with how intelligent agents ought to take actions in an environment in order to maximize the notion of cumulative reward.

- Example: Gaming, Autonomous helicopter, Recommender Systems

What kind of learning algorithms will apply for below applications

1. Given email labeled as spam/not spam, learn a spam filter.
2. Given a set of news articles found on the web, group them into set of articles about the same story.
3. Given a database of customer data, automatically discover market segments and group customers into different market segments
4. Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.

Time Table Slots: Machine Learning

Theory Hours:

Linear Regression

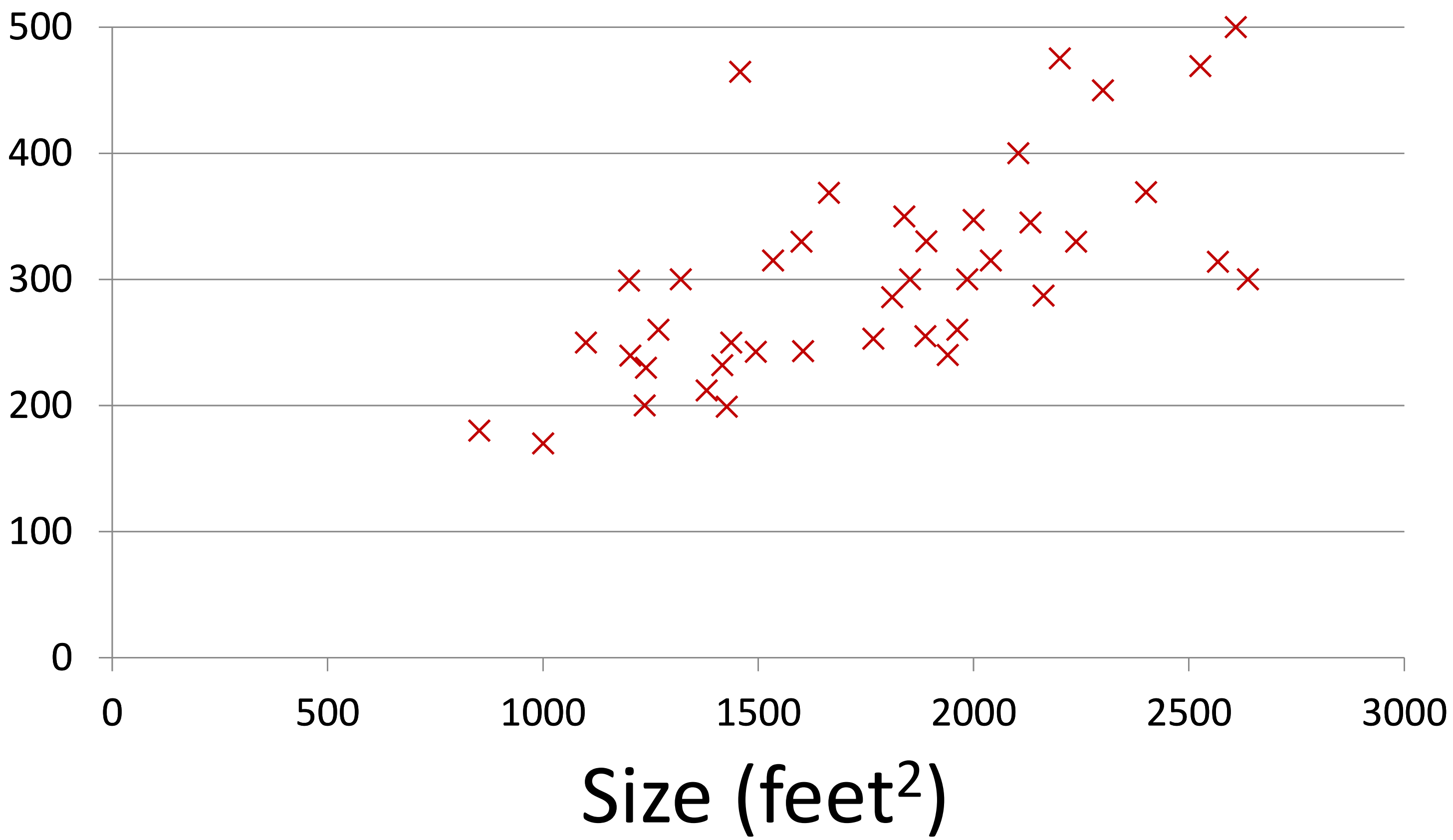
Training set of housing prices (Portland, OR)

Size in feet^2 (x)	Price ($) in 1000's (y)
2104	460
1416	232
1534	315
852	178
…	…

Notation:

m = Number of training examples
x's = "input" variable / features
y's = "output" variable / "target" variable

Housing Prices (Portland, OR)



<u>Supervised Learning</u>

Given the "right answer" for each example in the data.

<u>Regression Problem</u>

Predict real-valued output

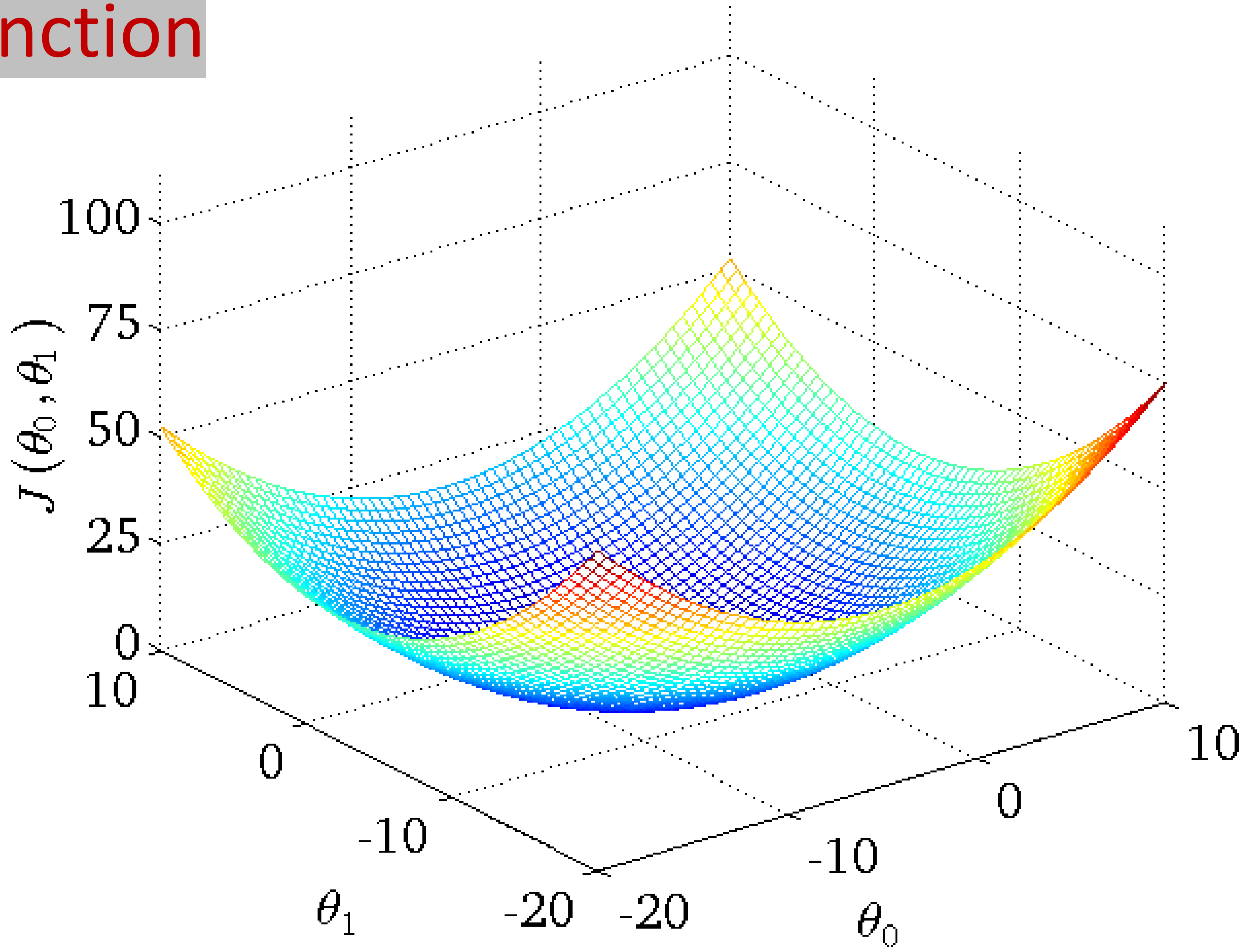
Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

Parameters: θ_0, θ_1

Cost Function: $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$

Goal: $\underset{\theta_0, \theta_1}{\text{minimize}}\ J(\theta_0, \theta_1)$

Cost Function



Multiple features (variables).

Size (feet^2) x	Price (\$1000) y
2104	460
1416	232
1534	315
852	178
…	…

$$h_\theta(x) = \theta_0 + \theta_1 x$$

Hypothesis: $h_\theta(x) = \theta^T x = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$

Parameters: $\theta_0, \theta_1, \ldots, \theta_n$

Cost function:

$$J(\theta_0, \theta_1, \ldots, \theta_n) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$$

Gradient descent:

Repeat $\{$

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \ldots, \theta_n)$$

$\}$ (simultaneously update for every $j = 0, \ldots, n$)